

The strongest association was found using a five-day moving average of ozone. No association was found in the September through April months. In co-pollutant models with a particulate matter or another gaseous pollutant, the ozone effect was only slightly diminished. The effects for PM and gaseous pollutants were generally significant in single pollutant models but diminished in co-pollutant models with ozone, with the exception of CO. The C-R functions for ozone are based on a single pollutant and two co-pollutant models, using the five-day moving average of one-hour max ozone.

Hospital Admissions, All Respiratory (ICD-9 codes 464, 466, 480-487, 493)

In a model with PM_{2.5}, the coefficient and standard error are based on the percent increase (33.0) and t-statistic (3.44) associated with a 45.2 ppb increase in the five-day moving average of one-hour max ozone (Burnett et al., 2001, Table 3).

G.2.2.2 Katsouyanni et al. (2009)

Katsouyanni et al. (2009) assessed the relationship between air pollution and hospital admissions from 1985 to 1994. Specifically, the authors examined hospitalizations due to respiratory diseases, conditions, or infections (ICD-9 codes 460-519) among U.S. citizens age 65 or older. The U.S. dataset included 14 cities with each city having data for 4 to 10 years from 1985-1994 and 7 cities having only summer ozone data. The investigators used a three-stage hierarchical model to account for within-city, within region, and between region variability. In the U.S. in adults 65 and older, authors found a .26% (.01, .51) increase in respiratory hospital admissions with 10 µg/m³ increase in ozone concentration using penalized splines with 8 df/year using lag 1. Authors found a .33% (.08, .58) increase in respiratory hospital admissions with 10 µg/m³ increase in ozone concentration using natural splines with 8 df/year using distributed lags.

G.2.2.3 Moolgavkar et al. (1997)

Moolgavkar et al. (1997) examined the relationship between air pollution and hospital admissions (ICD-9 codes 490-496) for individuals 65 and older in Minneapolis-St. Paul, Minnesota, from January 1986 to December 1991. In a Poisson regression, they found no significant effect for any of the pollutants (PM₁₀, ozone, or CO). The effect for ozone was marginally significant. The model with a 100 df smoother was reported to be optimal (p. 368). The health impact function for chronic lung disease is based on the results from a three-pollutant model (ozone, CO, PM₁₀) using the 100 df smoother; the function for Pneumonia uses the 130 df smoother.

Hospital Admissions, Chronic Lung Disease (ICD-9 codes 490-496)

In a model with CO and PM₁₀, the estimated coefficient and standard error are based on the percent increase (4.2) and 95% confidence interval of the percent increase (-1.0-9.4) associated with a change in daily average ozone levels of 15 ppb (Moolgavkar et al., 1997, Table 4).

Hospital Admissions, Pneumonia (ICD-9 codes 480-487)

In a model with NO₂, PM₁₀, and SO₂, the estimated coefficient and standard error are based on the percent increase (5.7) and 95% confidence interval of the percent increase (2.5-8.9) associated with an increase in daily average ozone levels of 15 ppb (Moolgavkar et al., 1997, Table 4).

G.2.2.4 Schwartz (1994a)

Schwartz (1994a) examined the relationship between air pollution and hospital admissions for individuals 65 and older in Minneapolis-St. Paul, Minnesota, from January 1986 to December 1989. In single-pollutant Poisson regression models, both ozone and PM₁₀ were significantly associated with pneumonia admissions. In a two-pollutant model, Schwartz found PM₁₀ significantly related to pneumonia; ozone was weakly linked to pneumonia. The results were not sensitive to the methods used to control for seasonal patterns and weather. The ozone C-R functions are based on the results of the single pollutant model and the two-pollutant model (PM₁₀ and ozone) with spline smoothing for temporal patterns and weather.

Hospital Admissions, Pneumonia (ICD-9 codes 480-487)

In a model with PM₁₀ and spline functions to adjust for time and weather, the coefficient and standard error are based on the relative risk (1.22) and 95% confidence interval (1.02, 1.47) for a 50 ppb increase in daily average ozone levels (Schwartz, 1994a, Table 4).

G.2.2.5 Schwartz (1994b)

Schwartz (1994b) examined the relationship between air pollution and hospital admissions (ICD codes 491-492, 494-496) for individuals 65 and older in Detroit, Michigan, from January 1986 to December 1989. In a two-pollutant Poisson regression model, Schwartz found both PM₁₀ and ozone significantly linked to pneumonia and COPD. The authors state that effect estimates were relatively unchanged compared to the unreported single pollutant models. No significant associations were found between either pollutant and asthma admissions. The C-R function for chronic lung disease incidence is based on the results of the “basic” co-pollutant model (ozone and PM₁₀) presented in Table 4 (p. 651). The study also reports results using generalized additive models to fit time and temperature variables, however no standard error or confidence intervals were reported.

Hospital Admissions, Chronic Lung Disease less Asthma (ICD-9 codes 490-492, 494-496)

The coefficient and standard error for the “basic” model are reported in Table 4 (Schwartz, 1994b, p.651) for a one ppb change in daily average ozone.

Hospital Admissions, Pneumonia (ICD-9 codes 480-487)

The ozone C-R function for pneumonia incidence is based on the coefficient and standard error for the “basic” co-pollutant model presented in Table 4 (Schwartz, 1994b, p. 651).

G.2.2.6 Schwartz (1995)

Studies have reported associations between short term changes in air pollution and respiratory hospital admissions. This relationship was examined in two cities with substantially different levels of sulphur dioxide (SO₂) but similar levels of airborne particles in an attempt to separate the effects of the two pollutants. Significant differences in weather between the two cities allowed the evaluation of that potential confounder also. Daily counts of admissions to all hospitals for respiratory disease (ICD 9 460-519) were constructed for persons aged 65 years and older in two cities - New Haven, Connecticut and Tacoma, Washington.

Each city was analysed separately. Average daily concentrations of SO₂, inhalable particles (PM₁₀), and ozone were computed from all monitors in each city, and daily average temperature and humidity were obtained from the US weather service. Daily respiratory admission counts were regressed on temperature, humidity, day of the week indicators, and air pollution. A 19-day weighted moving regression filter was used to remove all seasonal and subseasonal patterns from the data. Possible U- shaped dependence of admissions on temperature was dealt with using indicator variables for eight categories each of temperature and humidity. Each pollutant was first examined individually and then multiple pollutant models were fitted. All three pollutants were associated with respiratory hospital admissions of the elderly. The PM₁₀ associations were little changed by control for either ozone or SO₂. The ozone association was likewise independent of the other pollutants. The SO₂ association was substantially attenuated by control for ozone in both cities, and by control for PM₁₀ in Tacoma. The magnitude of the effect was small (relative risk 1.06 in New Haven and 1.10 in Tacoma for a 50 micrograms/m³ increase in PM₁₀, for example) but, given the ubiquitous exposure, this has some public health significance. The authors concluded that air pollution concentrations within current guidelines were associated with increased respiratory hospital admissions of the elderly. The strongest evidence for an independent association was for PM₁₀, followed by ozone.

Hospital Admissions, All Respiratory (ICD-9 codes 460-519) -- Tacoma

In a model with PM₁₀, the coefficient and standard error are estimated from the relative risk (1.20) and 95% confidence interval (1.06-1.37) for a 50 µg/m³ increase in average daily ozone levels (Schwartz, 1995, Table 6, p. 535). To calculate the coefficient, a conversion of 1.96 µg/m³ per ppb was used, based on a density of ozone of 1.96 grams per liter (at 25 degrees Celsius).

Hospital Admissions, All Respiratory (ICD-9 codes 460-519) -- New Haven

In a model with PM₁₀, the coefficient and standard error are estimated from the relative risk (1.07) and 95% confidence interval (1.00-1.15) for a 50 µg/m³ increase in average daily ozone levels (Schwartz, 1995, Table 3, p. 534). To calculate the coefficient, a conversion of 1.96 µg/m³ per ppb was used, based on a density of ozone of 1.96 grams per liter (at 25 degrees Celsius).

G.2.3 Emergency Room Visits

Table G-13 summarizes the additional health impacts functions used to estimate the relationship between ozone and emergency room (ER) visits. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-13. Additional Health Impact Functions for Ozone and Emergency Room Visits

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form
Asthma	Glad et al.	2012	Pittsburgh, PA	0-99		D8HourMax	0.003057	0.0011709	Logistic
Asthma	Ito et al.	2007	New York, NY	0-99		D8HourMax	0.005213	0.000909	Log-linear
Asthma	Ito et al.	2007	New York, NY	0-99	PM _{2.5}	D8HourMax	0.003976	0.000979	Log-linear
Asthma	Mar and Koenig	2009	Seattle, WA	18-99		D8HourMax	0.007696	0.0028374	Log-linear
Asthma	Mar and Koenig	2009	Seattle, WA	0-17		D8HourMax	0.010436	0.0043576	Log-linear
Asthma	Peel et al.	2005	Atlanta, GA	0-99		D8HourMax	0.000870	0.000529	Log-linear
Asthma	Sarnat et al.	2013	Atlanta, GA	0-99		D8HourMax	0.001113	0.0002828	Log-linear
Asthma	Wilson et al.	2005	Portland, ME	0-99		D8HourMax	0.003000	0.001000	Log-linear
Asthma	Wilson et al.	2005	Manchester, NH	0-99		D8HourMax	-0.001000	0.002000	Log-linear

G.2.3.1 Glad et al. (2012)

Glad et al. (2012) is summarized in section G.1.4.1.

Emergency Room Visits, Asthma

Using a case-crossover methodology, which controls for the effects of subject-specific covariates such as gender and race, a 2.5% increase was observed in asthma ED visits for each 10 ppb increase in the 1-hour maximum ozone level on day 2 (odds ratio [OR] = 1.025, p < 0.05).

G.2.3.2 Ito et al. (2007)

The authors assessed the effect of multi-collinearity among gaseous co-pollutants of PM_{2.5} and weather variables. The authors compiled daily meteorological and pollutant data in New York City for the years 1999 to 2002 and analyzed the effect of ozone on